SWR and Power Meter with Wi-Fi Interface

Setup and operating instructions · firmware of 2026-10-16

Specifications

Power supply	microUSB, 5 V, 1 A
Frequency range	1–30 MHz
Full-scale power	1000 W (a 2000 W firmware version is available)
Measured power	PEP (Peak Envelope Power)
SWR measurement	from ~10 W (from ~20 W on the 2000 W scale)
Wi-Fi range	up to 30–50 m with an entry-level router
Housing	Gainta G0479
Module	NodeMCU 0.9 (ESP8266)

Any modern browser on a computer, tablet or smartphone will do (Chrome, Firefox, Safari, Edge).

Device access point

Network name (SSID)	**SWRmeter_XXXXXX**, where XXXXXX is the unique chip ID, different for every device
Password	**1234567890**
IP address	**192.168.4.1**

Device pages:

- `http://<IP>/` — start page;
- `http://<IP>/swr` — SWR and power meter;
- `http://<IP>/wifi` — Wi-Fi settings, connection status and uptime.

If the device runs the firmware with authentication, the browser asks once for a login and password: **admin** / **swr12345**.

First connection to your Wi-Fi network

> Before you start, make sure DHCP is enabled on your router.

1. Connect the power. The LED on the module is on: the device is not yet connected to your network and works as an access point.
2. On your smartphone or computer, find the **SWRmeter_XXXXXX** network and connect to it with the password **1234567890**. On many devices the setup page opens automatically.
3. If it does not, open **http://192.168.4.1/wifi** in the browser.
4. Enter your network name (available networks are listed on the same page) and password, then click **Connect**.
5. The device connects to the router, the LED turns off, and the SWRmeter_XXXXXX access point is switched off.
6. Find the IP address the router gave to the device (in the router's client list) and open **http://<IP>/swr** in the browser. It is convenient to reserve a fixed address for the device in the router's DHCP settings.

The network name and password are stored in memory and survive power-off. If the network is lost, the device turns its access point back on (the LED lights up) and tries to reconnect every 30 seconds. To change the network, connect to the access point and repeat steps 3–4.

Meter page

- **Power** — peak envelope power (PEP) in watts. Up to 100 W it is shown in 1 W steps, above that in 5 W steps, above 1000 W in 10 W steps.
- **The bar** shows power relative to full scale: green up to 50 %, yellow up to 80 %, red above. The white marker is the maximum over the last 2 seconds.
- **SWR** is measured at the power peak. During pauses the last value is held for 3 seconds, then “---” is shown. “>10” means a very high SWR (open or short in the antenna).
- After transmission ends, the readings quickly return to zero.

The **SSB / CW / FT8** buttons select how the readings behave:

Mode	Behaviour
SSB	Readings are held for 0.7 s, so power is readable on speech instead of jumping with every syllable.
CW	Readings are held for 0.3 s to bridge the gaps between dots and dashes.
FT8	For constant-carrier modes (FT8, FT4, RTTY, AM, FM, tuning): fast rise to the level and light smoothing.

The selected mode is kept until the device restarts; after power-on it is always SSB.

Troubleshooting

Symptom	What to do
Values on the page dim, power drops to 0	The browser has received no data from the device for more than 2 seconds, i.e. the Wi-Fi link is lost. The readings become bright again when the link recovers. If this happens only while transmitting, see below.
The page does not open	Check that the phone or computer is on the same network as the device and that its IP address has not changed. If the LED is on, the device is not connected to the router: connect to the SWRmeter_XXXXXX network and open 192.168.4.1/wifi.
Does not connect to the router	Check the network password and that the router works in the 2.4 GHz band (the module does not support 5 GHz).
SWR shows “---”	Power is too low to measure SWR (below ~10 W) or there is no transmission.

RF interference

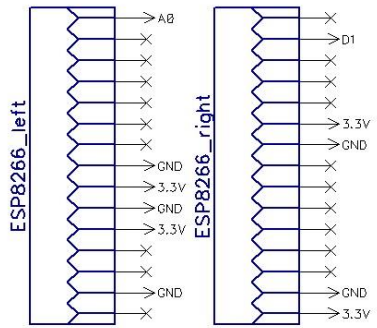
If the readings dim while transmitting, especially at high power or with elevated SWR, and the uptime (on the / wifi page) is not reset, RF from the transmitter is disturbing the Wi-Fi module. The device itself does not restart, but loses its link to the router for a few seconds. You can check this with `ping <device IP>` during transmission: there will be packet loss and delays of 1–2 seconds.

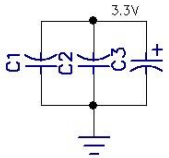
With elevated SWR, common-mode currents flow on the coax braid and the cable next to the meter starts to radiate. What helps, in order of effectiveness:

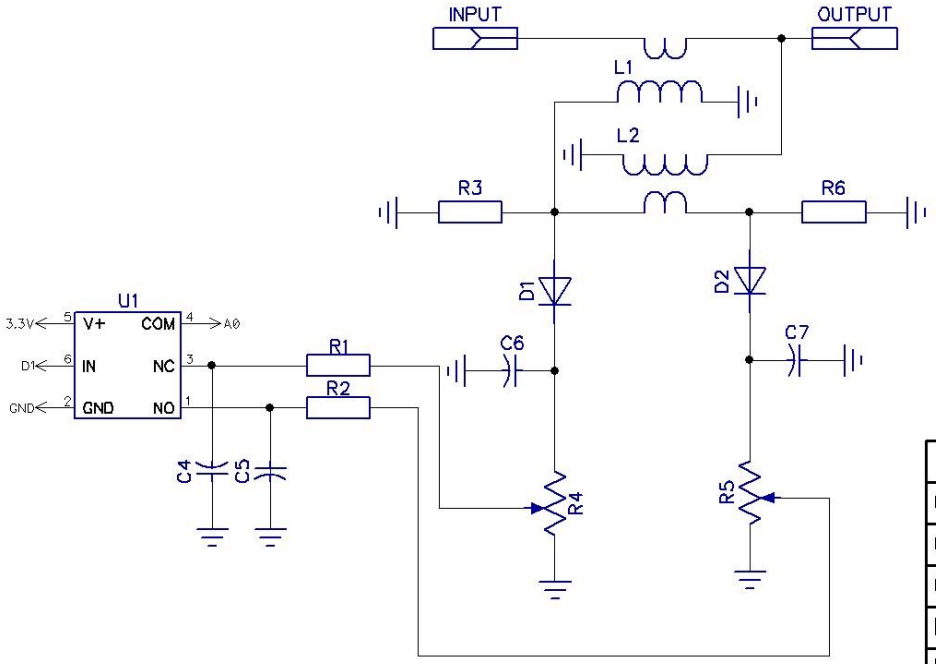
1. A common-mode choke on the cable right after the meter on the antenna side: 5–7 turns of coax on a ferrite toroid (mix 31 or 43) or several clamp-on ferrites.
2. A ferrite toroid on the USB power cable at the housing, 3–5 turns.
3. Matching the antenna: with SWR close to 1 interference is much lower.
4. Placing the meter away from the antenna and from radiating sections of the cable.

The schematic is on the next page. Source code, ready firmware and detailed description: https://github.com/ra0sms/SWR_PWR_meter · https://ra0sms.com/

Power and SWR meter with WI-FI interface







L1,L2 - 30 turns core FT82-43

RefDes	Value	Name
C1,C6,C7	10n	SMD0805
C2,C4,C5	100n	SMD0805
C3	220u	16V
D1,D2	1N5711	
R1,R2	1k	SMD0805
R3,R6	51	2W
R4,R5	10k	3296W
U1	TS5A3159	